Assignment 2: Visualization by example

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Exercise 1

```
library(tidyverse)
library(palmerpenguins)
```

Exercise 2

i. This dataset has 344 rows and 8 columns.

ii. The unit of observation is penguins.

iii. Three categorical variables in the penguins dataset are species, island, and sex.

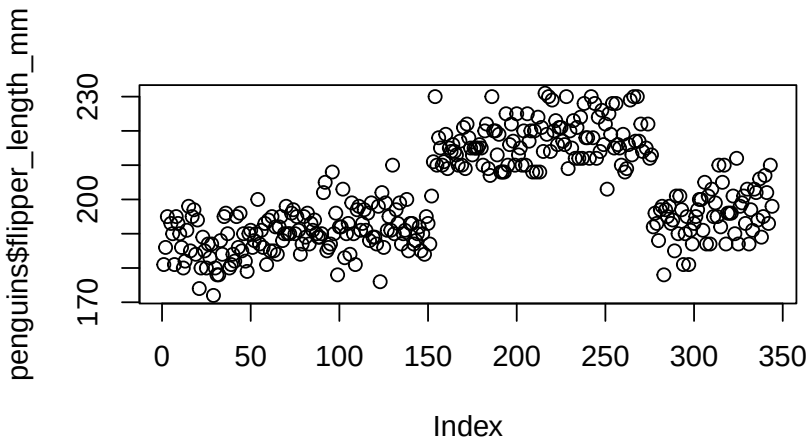
iv. Four continuous variables in the penguins dataset are bill length, bill depth, flipper length, and body mass.

v. The year could be treated as either continuous or categorical, depending on the context in which it is used.

vi. The three species of penguin in the dataset are Adelie, Gentoo, and Chinstrap.

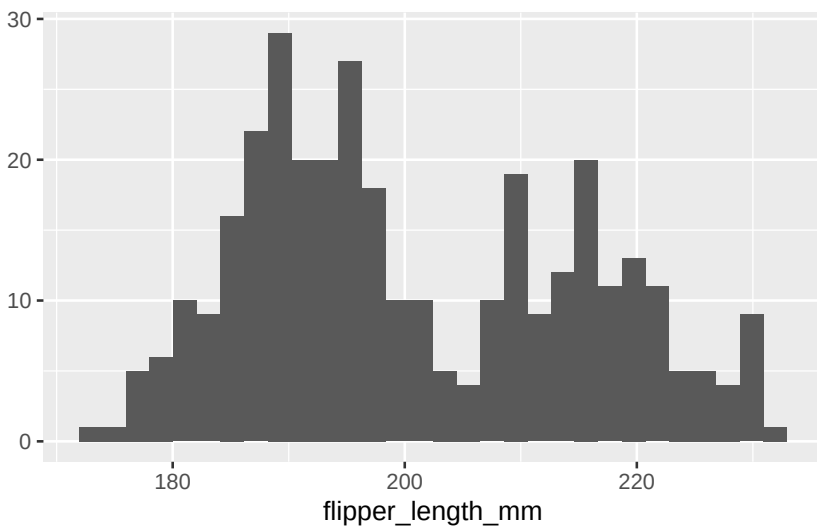
Exercise 3

```
plot(x = penguins$flipper_length_mm)
```



Exercise 4

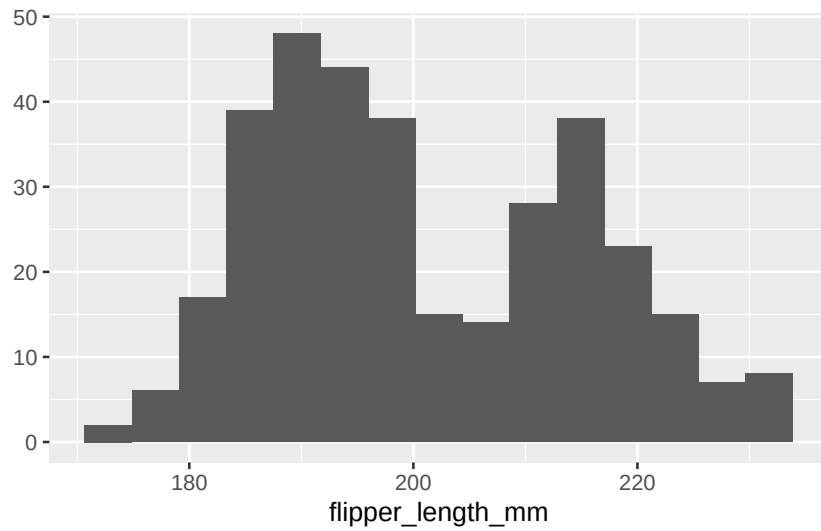
```
qplot(x = flipper_length_mm, data = penguins)
```



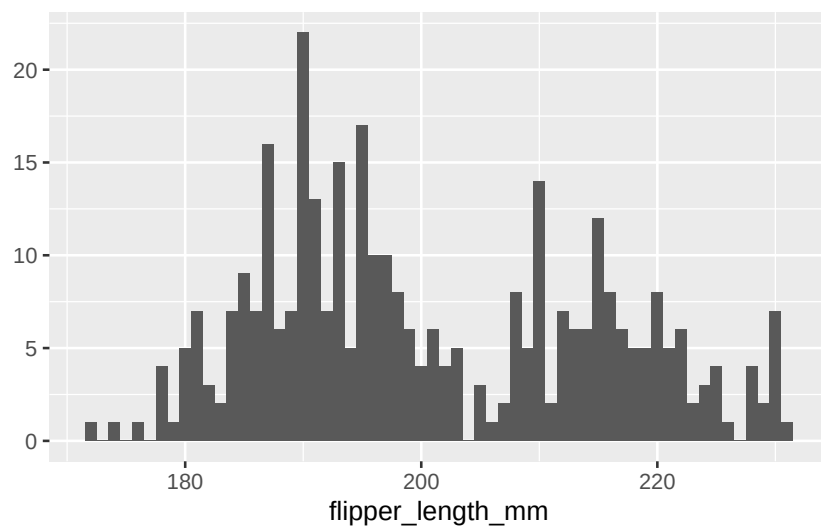
ii. The first graph is a scatterplot while the second graph is a histogram. The scatterplot does not compare two variables; rather, it shows the distribution of flipper length in relation to each penguin. The histogram, on the other hand, indicates how flipper length falls into three distinct bins (180, 200, and 220). The scatterplot uses the operator $ to first reference the dataset name then pull the flipper length. The histogram uses the operator =, first indicating flipper length then where the data comes from (penguins).

Exercise 5

```r
qplot(x = flipper_length_mm, bins = 15, data = penguins)
```



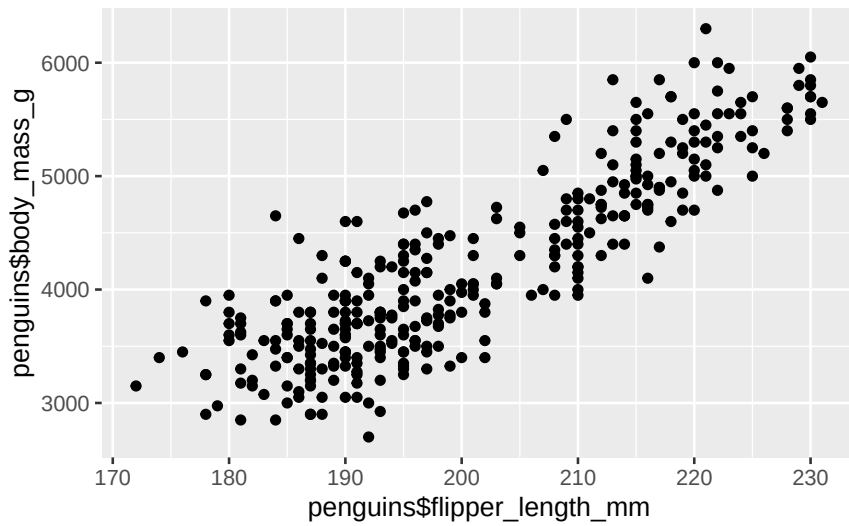

```r
qplot(x = flipper_length_mm, binwidth = 1, data = penguins)
```



iii. While bins simply indicates the number of bins, binwidth indicates the range of values in each bin. It is important to experiment with different binwidth values to see which conveys the clearest message to the audience. Changing these values affects how cluttered or smooth the data appears.

Exercise 6

```
qplot(x = penguins$flipper_length_mm, y = penguins$body_mass_g)
```

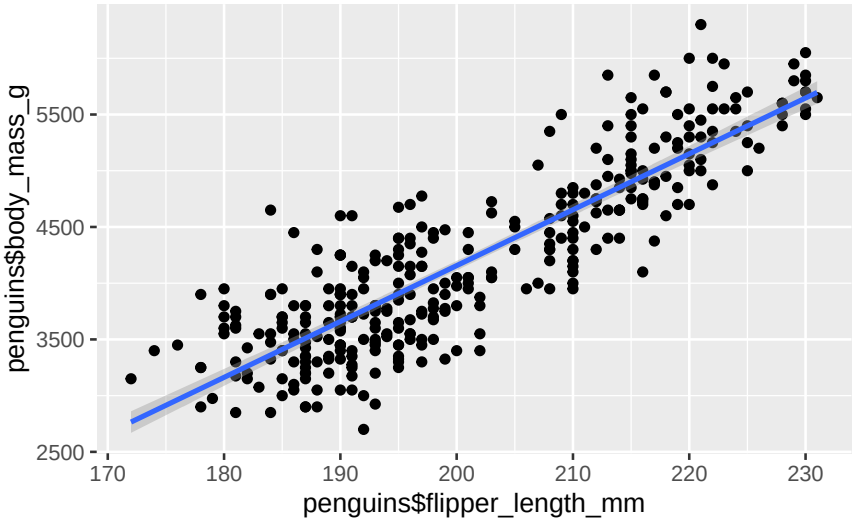


ii. The variable on the y-axis is body mass.

iii. There appears to be a linear relationship between these two variables.

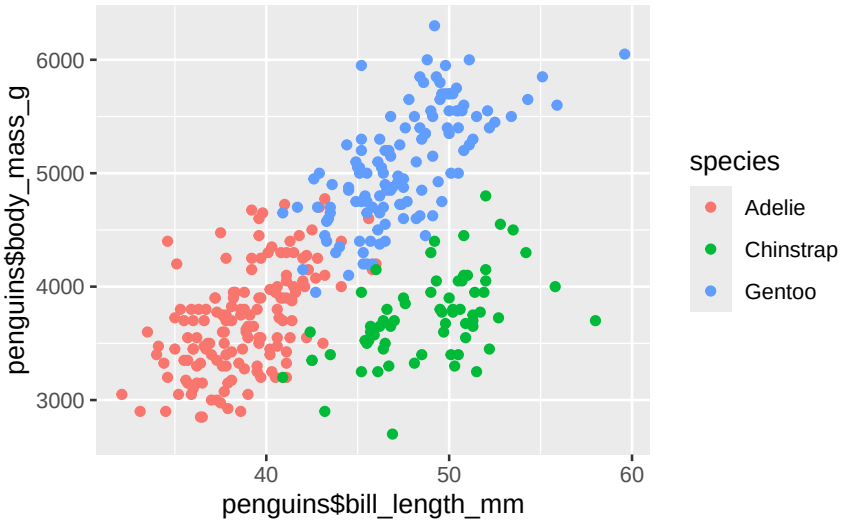
Exercise 7

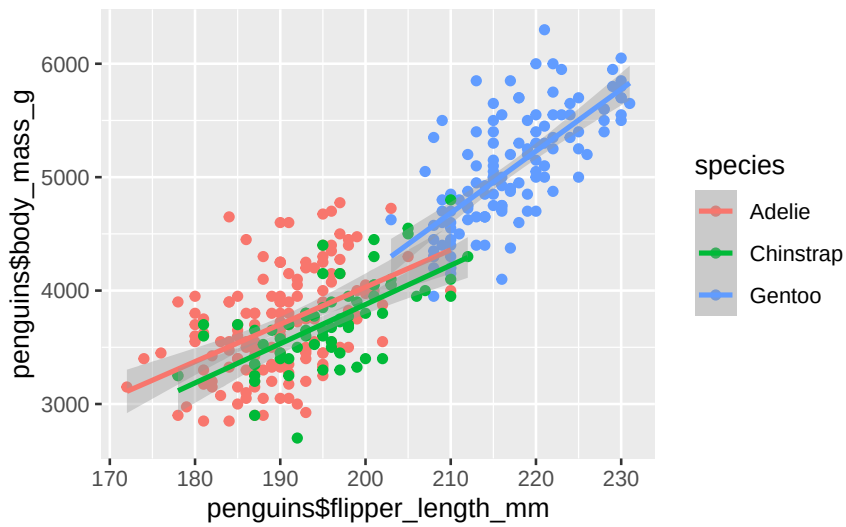
```
qplot(
   x = penguins$flipper_length_mm,
   y = penguins$body_mass_g,
   data = penguins,
   geom = c("point", "smooth"),
   method = "lm"
)
```



Exercise 8

```
qplot(
x = penguins$bill_length_mm,
y = penguins$body_mass_g,
color = species,
geom = "point",
data = penguins
)
```



Exercise 9

```r
qplot(
  x = penguins$flipper_length_mm,
  y = penguins$body_mass_g,
  color = species,
  geom = c("point", "smooth"),
  method = lm,
  data = penguins
)
```